PVP 14

Code: AE1T4, EC1T5

I B.Tech - I Semester – Regular Examinations – November 2015

C PROGRAMMING
(Common to AE, ECE)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1. a) What is the difference between pointers and arrays?
   b) Write about Enumerated data types.
   c) Explain advantages of recursion.
   d) Distinguish between getchar and putchar.
   e) What are the different storage classes in C?
   f) Explain strstr.
   g) Explain the features of structures.
   h) Write a program to check equivalence of two numbers. Use "if" statement.
   i) Explain fseek( ), ftell( ).
   j) Explain about type casting.
   k) Write about volatile variables.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks. 3 x 16 = 48 M

2. a) Define Algorithm and flow chart. 8 M

   b) Write an algorithm and draw the flow chart for finding the biggest among the three given numbers. 8 M

3. Explain different types of arrays with an example. 16 M

4. a) Write a C function to find GCD of two numbers using recursion. Call this function in a main program. 8 M

   b) Give a brief note about different storage classes. 8 M

5. Explain Predefined Macros. 16 M

6. a) How Union is differ from structure? 8 M

   b) What are steps for file operations? 8 M

PVP 14

Code: EC1T5, AE1T4

I B. Tech - I Semester – Regular/Supplementary Examinations
December 2016

C - PROGRAMMING
(Common for ECE & AE)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks
11 x 2 = 22 M

1.
a) What is flow chart?
b) List the functional units of computer.
c) Write the syntax of printf() and scanf()
d) What happens if the break is not used in the switch case statements?
e) Write any two differences of while and do while.
f) How can we declare and initialize a multi dimensional array?
g) Define recursion.
h) Define the term scope of a variable.
i) What is a null pointer?
j) Differentiate structure and union.
k) List different storage classes.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.
3 x 16 = 48 M

2. a) Explain the Structure of C program. 8 M

b) Write briefly about the C Tokens with suitable examples. 8 M

3. a) What are the different types of control statements available 'C'? Explain them with an example. 12 M

b) Write a program to find the reverse of a given integer using while. 4 M

4. a) Define String and Explain Various String Manipulation Functions. 10 M

b) Write a program for reversing a string without using Library function. 6 M

5. a) Distinguish between array of pointers and pointer to array with examples. 8 M

b) Explain pointer to function and function returning pointer with example. 8 M

6. a) Describe structure declaration, initialization and accessing elements. 8 M

b) Write a program to copy one file data into another file. 8 M

PVP 14

Code: EC1T5, AE1T4

I B. Tech - I Semester – Regular/Supplementary Examinations
November 2017

C - PROGRAMMING
(Common for ECE & AE)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1.

a) Draw a flow chart for finding sum of **n** numbers.

b) Consider the following program

```
#include<stdio.h>
main()
{
        int a=5;
        b = a << 2;
        printf("a=%d   b=%d \n", a,b);
}
```

What is the output of above program? Explain it in two lines.

c) Write the syntax of switch-case. Write the role of default case in it.

d) Consider the following program

```
#include<stdio.h>
main()
{
```

```
        int i;
        for(i=0; i<10 ; i += 2)
        printf("%d   ", i);
}
```

What is the output of above program? Explain it in two lines.

e) What is the use of **static** keyword in C. Give an example for it.

f) Write a C program to compute **gcd** of two numbers using recursion.

g) How to declare a pointer to a pointer and write a program which uses this concept.

h) Consider the following program

```
#include<stdio.h>
#define DEF
main()
{
        #ifdef DEF
                printf("Hello\n");
        #else
                printf("Hai");
        #endif
}
```

What is the output of above program? Explain it in two lines.

i) Define Array with an example.

j) What are bit fields? Give an example for it.

k) What do you know about environment variables?

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2.a) Explain about different data types in C language. 8 M

b) Write an algorithm to find smallest of three given numbers. 8 M

3.a) Write a C program to find multiplication of two matrices. 8 M

b) Write a C program to read **n** strings and sort them in ascending order**.** 8 M

4.a) Write about different types of functions in terms of arguments and return types. 8 M

b) What are different parameter passing mechanisms exists in C? Explain each method with an example program. 8 M

5.a) Define void pointer. Where we use this concept? Give an example for it. 8 M

b) Write briefly about standard I/O predefined streams in **stdio.h**. 8 M

6.a) Define a structure with the name **student**. Assume appropriate fields in student structure. Develop a program which reads **n** students data and displays all **n** students information. 8 M

b) Write about different built-in functions used in Files concept**.** 8 M

PVP 14

Code: EC1T5, AE1T4

I B.Tech - I Semester – Regular/Supplementary Examinations
November 2018

C - PROGRAMMING
(Common for ECE, AE)

Duration: 3 hours Max. Marks: 70

PART – A

Answer ***all*** the questions. All questions carry equal marks

11 x 2 = 22 M

1. a) What is an algorithm? Write an algorithm to swap two numbers.

   b) Write a small program to explain comma operator.

   c) What is an array? How it is different from a normal variable.

   d) Explain strcat( ) and strcmp( ) functions in C.

   e) Explain scope of a variable with an example.

   f) Write a program to add two numbers using pointers.

   g) What is a macro? Write a program to explain the concept.

   h) Define structure. Explain with an example.

   i) How files are useful in C. Write syntax for opening and closing a file.

   j) Write a program to print Fibonacci series using recursion.

   k) Write advantages of Recursion.

PART – B

Answer any ***THREE*** questions. All questions carry equal marks.

3 x 16 = 48 M

2. a) Explain the following
   - i) Machine language 2 M
   - ii) Assembly language 3 M
   - iii) High level language 3 M

   b) Explain the difference between type conversion and type casting with suitable examples. 8 M

3. a) Explain various loop control structures in C with an example . 8 M

   b) Write a program two multiply two matrices. 8 M

4. a) What is a function? Explain different types of functions with suitable examples. 8 M

   b) Explain various storage classes in C with examples. 8 M

5. a) Explain different ways of allocating memory in C. Write a program to illustrate the functions associated with memory allocation. 8 M

b) i) Explain how macro is different from function concept in C. 3 M

ii) What is the role of Preprocessor in the Compilation process and explain two preprocessor directives. 5 M

6. a) What are the different ways to access the members of structure elements in C. Give example for each case. 8 M

b) Explain various 4 file handling methods with their syntax. 8 M